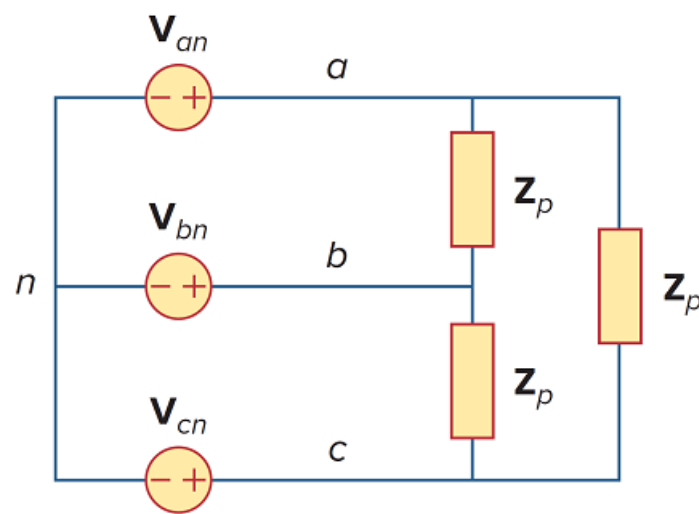


Problem

If wattmeters W_1 and W_2 are properly connected respectively between lines a and b and lines b and c to measure the power absorbed by the delta-connected load in Fig. 12.44, predict their readings.

Figure 12.44



Step-by-step solution

Step 1 of 1

From problem 12.11

$$V_{AB} = 220 \angle 130^\circ \text{ V} \quad \& \quad I_{aA} = 30 \angle 180^\circ \text{ A}$$

$$P_1 = (220)(30) \cos(130^\circ - 180^\circ) = 4242 \text{ W}$$

$$V_{CB} = -V_{BC} = 220 \angle 90^\circ$$

$$I_{cC} = 30 \angle -60^\circ$$

$$P_2 = (220)(30) \cos(190^\circ + 60^\circ)$$

$$= -2257 \text{ W} .$$